1 Warm-up

1. Last time we argued that the non-existent integral

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} dx$$

is supposed to give the arc length of the unit semicircle. Find a way to make sense of this integral. Explain your proposition and show that it gives the correct answer.

2 Improper Integrals of Type 2: Unbounded Integrand

Definition 2.1 (Improper Integrals of Type 2)**.** (a) If f is continuous on $[a,b)$ and is discontinuous at b, then define

$$\int_a^b f(x)dx := \lim_{t\to b^-}\int_a^b f(x)dx,$$

provided this limit exists (as a finite number).

(b) If f is continuous on $(a,b]$ and is discontinuous at a, then define

$$\int_a^b f(x)dx := \lim_{t\to a^+}\int_t^b f(x)dx,$$

provided this limit exists (as a finite number).

The improper integral $\int_a^b f(x)dx$ is called *convergent* if the corresponding limit exists and *divergent* if the limit does not exist.

(c) If f has a discontinuity at c, where $a<c<b$, and both $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ are convergent, then we define

$$\begin{aligned}\int_a^b f(x)dx :&= \int_a^c f(x)dx + \int_c^b f(x)dx\\ &= \lim_{s\to c^-}\int_a^s f(x)dx + \lim_{t\to c^+}\int_c^b f(x)dx.\end{aligned}$$

2. Find $\displaystyle\int_2^5 \frac{1}{\sqrt{x-2}}dx$.

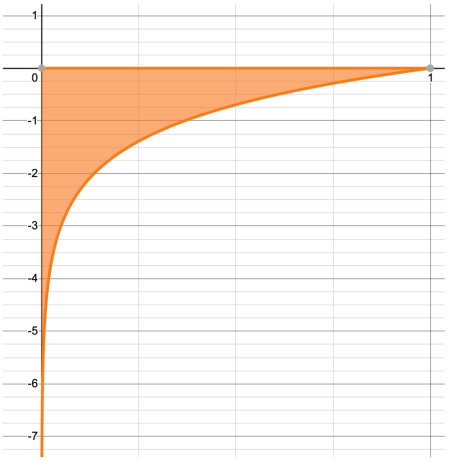
3. Determine whether $\int_0^{\pi/2} \sec(x)dx$ converges or diverges.

4. Explain what is wrong with the following calculation of the following integral:

$$\int_0^3 \frac{dx}{x-1} = [\ln|x-1|]_0^3 = \ln(2) - \ln(1) = \ln(2).$$

Evaluate $\int_0^3 \frac{dx}{x-1}$ correctly, or show that it diverges.

5. Determine whether the region in the plane bounded by the coordinate axes and the graph of $\ln x$ has finite area.



6. For what values of p is the integral

$$\int_0^1 \frac{1}{x^p} dx$$

convergent?